

CALCULUS I, Final Exam, 29/04/26

Duration: 3 hours, total marks: 80

1. (15 points) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a C^2 function such that $f(\mathbb{R}^2)$ is a bounded set in $\mathbb{R}$ and $f(u) = f(v)$ whenever $|u| = |v|$. Let (x, y) denote the rectangular coordinates and let $(r = \sqrt{x^2 + y^2}, \theta)$ denote the polar coordinates on $\mathbb{R}^2$. By the hypothesis, we can write $f(x, y) = g(r)$ for a function $g : [0, \infty) \rightarrow \mathbb{R}$.

(a) Show that $D_{xx}f + D_{yy}f = g''(r) + \frac{1}{r}g'(r)$. Here $D_{xx}f$ denotes the second partial derivative $\frac{\partial^2 f}{\partial x^2}$.

(b) Suppose that $D_{xx}f + D_{yy}f = 0$ on $\mathbb{R}^2$. Show that $f(u) = f(v)$ for all $u, v \in \mathbb{R}^2$.

2. (25 points) Make the identification $\mathbb{C} \cong \mathbb{R}^2$. Fix an integer $n \geq 1$. Let $z_n, z_{n-1}, \dots, z_1, z_0 \in \mathbb{C}$ with $z_n \neq 0$. Consider the map

$$\phi : \mathbb{C} \rightarrow \mathbb{C}, \quad \phi(z) = z_n z^n + z_{n-1} z^{n-1} + \dots + z_1 z + z_0.$$

(a) If $B \subset \mathbb{C}$ is bounded, then show that $\phi^{-1}(B)$ is bounded.

(b) Let $(u_k)_{k \geq 1}$ be a convergent sequence in $\mathbb{C}$ with $u_k \rightarrow u$. If for every $k \geq 1$, $u_k = \phi(v_k)$ for some $v_k \in \mathbb{C}$, show that $u = \phi(v)$ for some $v \in \mathbb{C}$.

(c) Using the Inverse Function Theorem, show that there exists an open set $U \subset \mathbb{C}$ such that $U \subset \phi(\mathbb{C})$.

(d) Let X be a closed subset of $\mathbb{C}$ which contains an open set. Suppose that the following set is finite:

non empty
 $\{z \in \mathbb{C} \mid \text{there exist sequences } (w_n) \subset X, (w'_n) \subset \mathbb{C} \setminus X \text{ such that } w_n \rightarrow z, w'_n \rightarrow z\}.$

Show that $X = \mathbb{C}$.

(e) Show that there exists $z \in \mathbb{C}$ such that $\phi(z) = 0$.

3. (20 points)

(a) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^∞ function^[1]. Show that there exist C^∞ functions $g_1, \dots, g_n : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$f(x) - f(0) = \sum_{i=1}^n x_i g_i(x) \quad \forall x = (x_1, \dots, x_n) \in \mathbb{R}^n.$$

(b) Let $f = (f_1, f_2, f_3) : \mathbb{R}^5 \rightarrow \mathbb{R}^3$ be a C^∞ function such that the rank of $f'(a)$ is equal to 3 for every $a \in \mathbb{R}^5$. Let $g : \mathbb{R}^5 \rightarrow \mathbb{R}$ be a C^∞ function such that $g(a) = 0$ whenever $f(a) = (0, 0, 0)$.

Show that for every $a \in \mathbb{R}^5$, there exists an open set $U \subset \mathbb{R}^5$ containing a and C^∞ functions $g_i : U \rightarrow \mathbb{R}$ for $1 \leq i \leq 3$ such that

$$g(x) = g_1(x)f_1(x) + g_2(x)f_2(x) + g_3(x)f_3(x) \quad \forall x \in U.$$

Hint for part (b): Use the Implicit Function Theorem and part (a).

4. (10 points) Determine whether the following statements are true or false, with justification:

(a) If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a function such that the directional derivatives $D_u f(a, b)$ exist for all unit vectors $u \in \mathbb{R}^2$ and all points $(a, b) \in \mathbb{R}^2$, then f is continuous.

(b) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a C^∞ function. Let $S = [0, 1] \times [0, 1] \subset \mathbb{R}^2$ and let $g : S \rightarrow \mathbb{R}$ be the restriction of f to S . If g has a local extrema at $(a, b) \in S$, then $f'(a, b) = 0$.

5. (10 points) Determine the local and global extrema of the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(a, b) = abe^{\frac{-a^2-b^2}{2}}$.

¹i.e., f is infinitely differentiable